# Practical No. 7

**Aim:** Study & Implementation of Clauses.

**Theory:**

Structured Query Language (SQL) is a powerful language used to manage and manipulate relational databases. One of the essential features of SQL is its **clauses**, which allow users to filter, sort, group, and limit data efficiently. SQL clauses simplify querying and enhance database performance by retrieving only the necessary records based on specific conditions.

**What is Clauses in SQL?**
SQL clauses are built-in functions that define specific conditions within an SQL statement to retrieve, update, or manipulate data from a database. These clauses work alongside **SELECT, UPDATE, DELETE, and INSERT** queries to refine results and ensure efficient data handling.

**Key Functions of SQL Clauses**

- Filter and retrieve specific records from a database.
- Group data based on certain attributes.
- Sort query results in ascending or descending order.
- Limit the number of records displayed in a query result.

SQL clauses are integral to writing optimized queries that return precise results without scanning unnecessary data.

**WHERE Clause:**
The WHERE clause is used to filter records based on specific conditions. It is typically used in SELECT, UPDATE, and DELETE queries to restrict the data that is affected by these statements.

**Example:** SELECT * FROM Students

WHERE stu_fees < 3500;

**ORDER BY Clause:**

The ORDER BY clause is used to sort the query results in either ascending or descending order. It is commonly used with numeric, date, and text fields to organize data meaningfully, such as sorting employees by their joining date.

**Syntax: SELECT <set of fields> FROM <relation_name>**

**ORDER BY <field_name>;**

**Example:** SELECT * FROM Students

ORDER BY stu_fees ASC;

**GROUP BY Clause:**

The GROUP BY clause groups records with the same values in specified columns and is used with aggregate functions like COUNT(), SUM(), AVG(), etc.

**Syntax: SELECT <set of fields> FROM <relation_name>**

**GROUP BY <field_name>;**

**Example:** SELECT stu_class, SUM(stu_fees) AS total_fees

FROM Students

GROUP BY stu_class;

**HAVING Clause:**

The HAVING clause is similar to WHERE but is used to filter grouped records. It is used with GROUP BY to apply conditions on aggregated results, such as filtering groups where the total revenue exceeds a certain amount.

**Syntax: SELECT column_name, aggregate_function(column_name) FROM table_name**

**WHERE column_name operator value**

**GROUP BY column_name**

**HAVING aggregate_function(column_name) operator value;**

**Uses of SQL Clauses**

SQL clauses have multiple applications in data filtering, sorting, and aggregation. Below are some common use cases:

**WHERE Clause:** Used to filter records that meet specific conditions, such as retrieving employees with a salary greater than 50,000.

**ORDER BY Clause:** Helps in sorting the retrieved data in either ascending or descending order, such as sorting students based on their marks.

**GROUP BY Clause:** Used to group records with identical values, often combined with aggregate functions like SUM() or COUNT(), for example, calculating total sales per region.

**HAVING Clause:** Applies conditions on grouped data, such as filtering groups where the total sales exceed 100,000

**Note: Write a query to be executed in lab with screen shots.**

**Result:**

**Thus, we execute the different types of clauses in SQL successfully.**